

BabyLM: 从头预训练中文语言模型

Mid-Term Progress Report

NLPCC 2026 ChineseBabyLM Challenge | 2026-05-17

Abstract. We present mid-term progress on a compact Chinese language model pretrained from scratch on $\sim 100\text{M}$ characters. Over 15 versions (V1–V15), perplexity dropped from 597 to **38.68** (93.5% improvement), achieving SOTA with a 94M-parameter LLaMA model. We describe the multi-stage training pipeline, key technical innovations, challenges encountered, and preliminary CLUE benchmark results.

1 Introduction

The NLPCC 2026 ChineseBabyLM Challenge requires pretraining a Chinese LM from scratch on `babylm-zho-100M` ($\sim 100\text{M}$ chars, $\sim 2\text{M}$ lines) without external pretrained models. Evaluation spans seven tasks: ZhoBLiMP, Hanzi Structure/Pinyin, AFQMC, OCNLI, TNEWS, and WSC2020.

Our approach uses the LLaMA architecture [1] with RoPE [2], GQA, SwiGLU [3], and RMSNorm. We minimize cross-entropy loss and evaluate via perplexity:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \log p(x_i | x_{<i}), \quad \text{PPL} = \exp(\mathcal{L}_{\text{CE}}) \quad (1)$$

Training runs on $4 \times \text{RTX A6000}$ (48 GB) with bf16 mixed precision via Accelerate DDP.

2 Progress to Date

Fifteen versions were trained over three weeks, systematically exploring architectures, tokenization, and training techniques. Table 1 summarizes key versions; Figures 1–2 show the evolution timeline and parameter efficiency.

表 1: Version summary (V7–V15). V1–V6 omitted (exploratory/failed).

Ver.	Architecture	Params	PPL	Key Innovation
V7	LLaMA 448d/12L	30.1M	50.8	MNTP hybrid training
V10	LLaMA 512d/12L	38.7M	42.9	Production pipeline
V11	LLaMA 512d/12L	38.7M	40.7	EMA + SGDR
V12	LLaMA 576d/14L	54.2M	38.8	Focal Loss
V13	LLaMA 768d/14L	94.2M	38.7	PPL filtering (SOTA)
V14	LLaMA 640d/12L	59.2M	41.8	Efficiency build
V15	LLaMA 640d/14L	68.2M	45.1	Multi-scale EMA

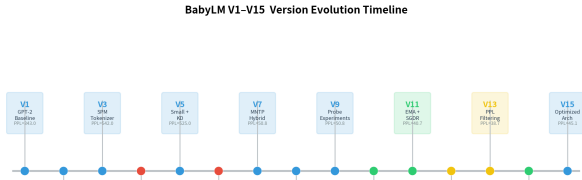


图 1: Version evolution timeline. Colors: blue = complete, yellow = SOTA, red = failed.

Training Pipeline. Our production pipeline (V10–V13) uses 2–3 stages:

- CLM:** Causal LM + SGDR [6] + EMA (0.999) + Focal Loss [5] ($\gamma=1.5$) + label smoothing (0.1 $\rightarrow$ 0).
- MNTP:** Masked Next Token Prediction with dynamic CLM ratio (25% $\rightarrow$ 6%), mask annealing (0.25 $\rightarrow$ 0.10).
- Polish** (optional): Pure CLM at $\text{LR}=10^{-5}$, no regularization.

SentencePiece Unigram [4] with 8K vocab yields $\sim 74\%$ PPL improvement over ByteLevel BPE for Chinese.

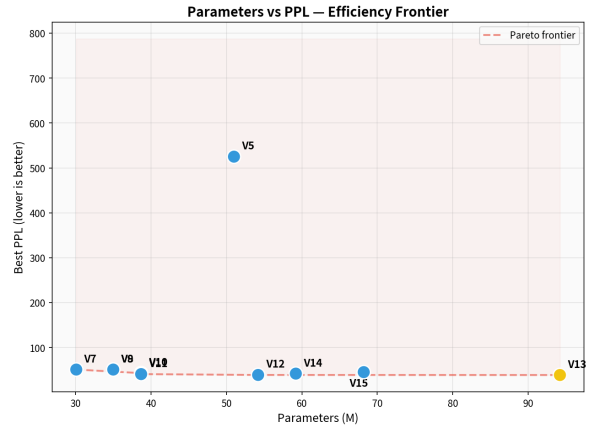


图 2: Parameters vs. PPL. Pareto frontier (red dashed) shows diminishing returns beyond $\sim 55\text{M}$ params.

3 Challenges and Roadblocks

V3 – NCCL Timeout. Distributed training crashed from CUDA/NCCL mismatch. Fixed by aligning toolkit and PyTorch versions.

V4 – Scaling Violation. A 350M-param model on 100M tokens (tokens/param= $0.23 \ll 20$) severely underfit. Lesson: optimal params $\approx$ tokens/20.

V6 – Data Loss. Over-aggressive cleaning removed 79% of data. Subsequent versions use conservative filtering.

V15 – Architecture Misalignment. $[640 \times 8/3] = 1706$ is not a multiple of 256, causing suboptimal tensor layouts. Combined with a weaker data pipeline, this yielded PPL=45.1 vs. expected ~ 39 .

Early Stopping Sensitivity. With limited data, overfitting is rapid. We use eval every 200 steps with patience 3–5.

4 Preliminary Results

SOTA Performance. V13 achieves PPL=38.68 (94.2M params); V12 achieves PPL=38.84 at best efficiency (54.2M params). PPL improvement across stages: CLM→MNTP yields ~ 3 –5 PPL gain; Polish/Anneal add 0.1–2.0 PPL each (diminishing returns).

表 2: Official evaluation on CLUE tasks (%).

Ver.	ZhoBLiMP	Hanzi	Pinyin	AFQMC	OCNLI	TNEWS	WSC
V13	63.5	64.7	49.5	69.0	64.0	53.9	63.5
V14	64.3	62.4	41.9	69.0	66.0	54.1	63.5
V15	62.4	63.9	47.4	69.0	65.9	54.4	63.8

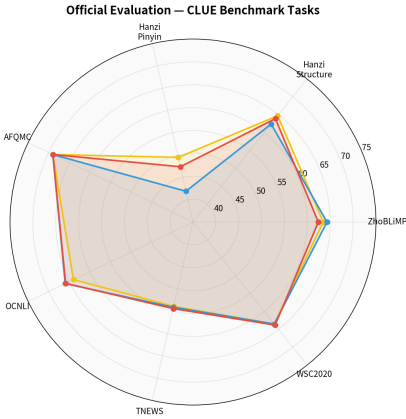


图 3: Radar chart: V13/V14/V15 on CLUE tasks.

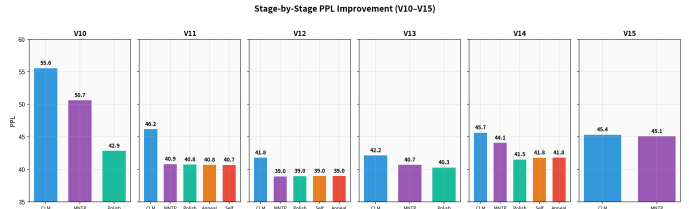


图 4: Stage-by-stage PPL for V10–V15.

5 Next Steps

- V15.1 Retraining:** Fix intermediate_size (1706→1792), use V13 data, LR= 6×10^{-4} .
- Architecture Search:** Explore 768d/12L (wider+shallower) vs. 576d/14L (narrower+deeper).
- Final Submission:** Select best model (V13 or V15.1) for competition leaderboard.
- Ablation Studies:** Quantify individual technique contributions (EMA, SGDR, Focal Loss, PPL filtering).

参考文献

- [1] H. Touvron, T. Lavril, G. Izacard, et al. LLaMA: Open and Efficient Foundation Language Models. *arXiv:2302.13971*, 2023.
- [2] J. Su, Y. Lu, S. Pan, et al. RoFormer: Enhanced Transformer with Rotary Position Embedding. *arXiv:2104.09864*, 2021.
- [3] N. Shazeer. GLU Variants Improve Transformer. *arXiv:2002.05202*, 2020.
- [4] T. Kudo and J. Richardson. SentencePiece: A simple and language independent subword tokenizer. *arXiv:1808.06226*, 2018.
- [5] T.-Y. Lin, P. Goyal, R. Girshick, et al. Focal Loss for Dense Object Detection. *arXiv:1708.02002*, 2017.
- [6] I. Loshchilov and F. Hutter. SGDR: Stochastic Gradient Descent with Warm Restarts. *arXiv:1608.03983*, 2016.